Mini Project Report

On

FarmHub – An Online Farm Website

Submitted By

K.Vishnu Kumar Reddy

R210701

Under the Guidance of

Mr. Satyanandaram N

MSIT (IIIT Hyderabad)

Lecturer

Computer Science and Engineering

Rajiv Gandhi University of Knowledge and Technologies

RGUKT RK Valley, Kadapa, 516330

Department of Computer Science and Engineering

2025-2026

Rajiv Gandhi University of Knowledge Technologies
RK Valley, Kadapa (Dist.), Andhra Pradesh - 516330

CERTIFICATE

This is to certify that the project report entitled "**FarmHub – An Online Farm Website**" being submitted by K.Vishnu Kumar Reddy (R210701) under my guidance and supervision and is submitted to DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING in partial fulfilment of requirements for the award of Bachelor of Technology in Computer Science and Engineering during the academic year 2025-2026 and it has been found worthy of acceptance according to the requirements of the University.

Project GUIDE	**HEAD OF THE DEPARTMENT**
Mr. Satyanandaram N	Dr. Ch. Ratna Kumari
MSIT (IIIT Hyderabad)	Mtech (UOH), Ph.D.(JNTU H)
Lecturer	Assistant Professor
RGUKT, RK VALLEY	RGUKT, RK VALLEY

EXTERNAL EXAMINER

DECLARATION

I hereby declare that this project work entitled **"FarmHub – An Online Farm Website"**, submitted to the **Department of Computer Science and Engineering**, is a genuine work carried out by me under the guidance of **Mr. Satyanandaram N**. This project is submitted in partial fulfillment of the requirements for the award of the degree of **Bachelor of Technology (B.Tech)**.

I further declare that this work has not been submitted elsewhere for the award of any other degree or diploma other than as specified above.

K.Vishnu Kumar Reddy
R210701

ACKNOWLEDGEMENT

I would like to express my sense of gratitude and respect to all those people behind the screen who guided, inspired, and helped me crown all my efforts with success. I express my gratitude to my project guide

Mr. Satyanandaram N *MSIT* for his valuable guidance at all stages of the study, advice, constructive suggestions, supportive attitude, and continuous encouragement, without which this project would not have been possible.

I extend my heartfelt thanks to **Dr. Ch. Ratna Kumari** *M.Tech, Ph.D.*, **Head of the Department, Computer Science and Engineering**, for her continuous support and for providing us with the resources and academic environment necessary for this project.

I sincerely acknowledge the academic support of **Mr. D. Ramesh Kailas,** *M.Tech* Dean of Academics, RGUKT RK Valley.

I am also grateful to **Prof. A. V. S. S. Kumara Swami Gupta** *M.Tech, Ph.D.* **Director of RGUKT RK Valley**, for his encouragement and for fostering a culture of research and innovation.

My sincere thanks go to the **faculty members of the Department of Computer Science and Engineering** for their guidance, and academic support.

Finally, I express my heartfelt gratitude to my parents for being a constant source of encouragement and inspiration, helping me keep my morals high.

With sincere Regards

K.Vishnu Kumar Reddy

R210701

Table of Contents

S.No	Content	Page No
1	Abstract	6
2 2.1 2.2	Introduction Problem Statement Objective	7 – 8 7 8
3	Technologies Used	8
4 4.1 4.2 4.3	App Architecture Core Modules Feature Modules Platform Specific Entry Points	9 – 14 14-16 16-17 17-18
5	Data Management and processing	18-20
6	Database Design	20-22
7	Proposed system workflow	22 – 25
8	Implementation Details	25 - 28
9	Challenges faced and solutions	28-30
10	Future Scope	30-33
12	Conclusion	46

Abstract:

In recent years, web-based technologies have significantly transformed the way people buy and sell products online. The increasing demand for convenience and accessibility has led to the development of e-commerce platforms across various domains, including agriculture. This project presents FarmHub – Farm Management System, an online web application designed to simplify the process of purchasing agricultural products through a digital platform. The system aims to provide users with an easy and efficient way to explore, select, and buy farm products from a centralized interface.

The FarmHub application is developed using modern web technologies and follows a structured approach to ensure usability and performance. The system allows users to register with a username and password, log in securely, and access various features such as viewing products, adding items to a cart, updating quantities, and placing orders. The interface is designed to be user-friendly and visually appealing, enhancing the overall user experience. Pagination and organized product display are implemented to handle multiple products efficiently.

The application also includes administrative functionalities that enable the admin to manage products, monitor user activity, and maintain the system effectively. Products can be added, updated, or removed through the admin panel, ensuring that the platform remains up-to-date. The integration of cart and order management features ensures a smooth transaction process for users.

FarmHub improves the traditional method of purchasing agricultural goods by offering a digital solution that saves time, reduces effort, and increases accessibility. It eliminates the need for physical presence and enables users to shop from anywhere at any time. The project demonstrates the practical implementation of an e-commerce system tailored for agricultural products, highlighting key concepts such as user authentication, database management, and web interface design.

Overall, FarmHub serves as an efficient and scalable platform that enhances the buying experience for users while providing a structured system for managing agricultural products online. This project showcases how web applications can be effectively utilized to modernize traditional systems and improve user interaction in the agricultural domain.

Introduction:

Problem Statement

With the rapid growth of digital technologies and online services, users increasingly prefer convenient and time-saving methods for purchasing products. However, in the agricultural sector, many people still rely on traditional methods of buying farm products, which involve physical visits to markets. This process can be time-consuming, less efficient, and sometimes inconvenient, especially for users who want quick access to fresh products.

Existing systems for purchasing agricultural products often lack proper organization, user-friendly interfaces, and real-time accessibility. Customers may face difficulties in exploring available products, comparing prices, and managing purchases efficiently. Additionally, there is often no centralized platform that allows users to view multiple farm products in one place and complete transactions smoothly.

To overcome these challenges, there is a need for a simple and efficient web-based system that enables users to browse, select, and purchase farm products online. The system should provide an easy-to-use interface, ensure smooth navigation, and support essential features like user authentication, cart management, and order processing.

Objective:

- To develop a web-based application that allows users to view and purchase agricultural products online.
- To design a user-friendly interface that enables easy navigation and enhances user experience.
- To implement a secure user authentication system using username and password for registration and login.
- To provide functionality for adding products to a cart, updating quantities, and managing selected items.
- To develop an order management system that allows users to place and track their orders efficiently.
- To create an admin panel for managing products, including adding, updating, and deleting product details.

- To ensure proper organization and display of products using features like pagination and categorization.
- To improve the overall efficiency of purchasing farm products by reducing time and effort through a digital platform.

Technologies Used:

The development of the FarmHub – Farm Management System involved a combination of frontend and backend technologies, along with database management tools to build a complete and functional web application. The system is designed to provide a smooth user experience and efficient handling of agricultural product data. The following technologies were used to develop and deploy the application

Frontend Development:

- **HTML (HyperText Markup Language):** Used to structure the web pages and design the layout of the application.
- **CSS (Cascading Style Sheets):** Used to style the application, improve visual appearance, and create a professional user interface.
- **Bootstrap (Optional):** Used for responsive design and to ensure the application works properly on different screen sizes.

Backend Development:

- **Django Framework:** A high-level Python web framework used to develop the backend of the application. It handles user authentication, routing, and overall system logic.
- **Python:** Used as the core programming language for implementing application functionality and business logic.

Database Management:

- **SQLite**: Used as the default database to store user details, product information, cart data, and order history.
- **Django ORM (Object Relational Mapping):** Used to interact with the database efficiently without writing complex SQL queries.

App Architecture:

The following section presents a comprehensive explanation of the core modules involved in the FarmHub system, detailing their internal workings, responsibilities, and interactions. The application is designed using a structured approach that separates frontend, backend, and database components to ensure scalability, maintainability, and efficient performance.

Core Modules:

The FarmHub – An Online Farm Website is built using multiple core modules that work together to provide a complete e-commerce experience for users. These modules handle user interaction, product management, cart operations, and order processing. Each module is designed independently to maintain modularity and simplify system development and maintenance.

1. User Management Module:

The User Management Module is responsible for handling all user-related activities such as registration, authentication, and session handling. It ensures secure access to the system and maintains user-specific data throughout the application lifecycle..

1.1. User Registration

- Allows users to create accounts using a username and password
- Validates user input to prevent invalid or duplicate entries
- Stores user credentials securely using encryption mechanisms
- Ensures proper data integrity and consistency in the database

1.2. User Login

- Authenticates users based on provided credentials
- Verifies username and password using Django authentication system
- Displays appropriate error messages for invalid login attempts
- Redirects users to the homepage/dashboard after successful login

1.3. Session Management

- Maintains user login sessions across multiple pages
- Enables personalized user experience during active sessions

- Restricts unauthorized access to protected pages
- Automatically logs out users after session expiration for security

2. Product Management Module:

This module manages all operations related to products available in the system. It ensures that users can easily browse, search, and view product details while allowing admins to manage product data efficiently.

2.1. Product Listing:

- Displays all available farm products to users
- Shows product details such as name, price, description, and image
- Provides a user-friendly interface for browsing products
- Supports dynamic data rendering from the database

2.2. Product Categorization:

- Organizes products into categories such as vegetables, fruits, grains, etc.
- Helps users quickly find relevant products
- Improves navigation and user experience
- Enables filtering and grouping of products

2.3. Admin Control:

- Allows admin to add new products into the system
- Enables updating of product details like price and description
- Provides functionality to delete products when required
- Manages product availability and stock status

3. Cart Management Module:

The Cart Management Module handles the temporary storage of selected products before final purchase. It acts as an intermediate stage between product selection and order placement.

3.1.Add to Cart

- Allows users to add selected products to the cart
- Stores selected items along with quantity
- Ensures smooth user interaction while shopping
- Updates cart dynamically without affecting other items

3.2. Update Quantity

- Allows users to increase or decrease product quantity
- Automatically updates total price based on quantity
- Prevents invalid quantities (like negative values)
- Improves flexibility in managing selected items

3.3. Remove Product from cart

- Enables users to remove unwanted products from the cart
- Updates cart contents instantly
- Provides better control over selected items
- Ensures clean and manageable cart interface

3.4. Cart Summary:

- Displays total items and total price
- Provides a clear overview before proceeding to checkout
- Helps users review their selections

- Enhances decision-making before purchase

4. Order Management Module

The Order Management Module handles the complete process of placing and managing orders. It ensures smooth conversion of cart items into orders and maintains records for tracking and reference.

4.1. Order Placement

- Converts selected cart items into a confirmed order
- Stores order details in the database
- Generates unique order records for each transaction
- Ensures secure and reliable order processing

4.2.Order Details:

- Stores complete product information including name, quantity, and price
- Maintains total cost and order summary
- Helps in accurate billing and record maintenance
- Provides detailed view of each order

4.3.Order History

- Displays previously placed orders for users
- Allows users to review past purchases
- Enhances transparency and user experience
- Maintains long-term transaction records

5. Admin Module:

The Admin Module provides control and management capabilities to the administrator of the system. It enables efficient handling of products, users, and orders through a centralized interface.

5.1. Product Control:

The admin can efficiently manage products.

- Enables admin to add new products
- Allows updating of product details such as price and description
- Provides option to delete unavailable or outdated products
- Ensures accurate and up-to-date product listings

5.2. Order Monitoring:

This ensures smooth order processing

- Allows admin to view all placed orders
- Tracks order status and progress
- Manages delivery-related information
- Maintains records of transactions for future reference

5.3. User Management:

The admin supervises user activities.

- Enables admin to view all registered users
- Allows monitoring of user activities
- Provides control to manage or restrict user accounts
- Handles misuse or unauthorized actions effectively

6. Authentication & Security:

This module ensures secure access and protection of user data within the system. It implements authentication and authorization mechanisms to maintain system integrity.

- Secure login mechanisms
- Password protection and encryption

- Role-based access control
- Prevention of unauthorized system usage
- Ensures confidentiality and data integrity

6.1. Secure Login:

- Verifies user credentials using authentication mechanisms
- Uses Django authentication system for validation
- Prevents unauthorized login attempts
- Provides error messages for invalid credentials
- Ensures safe and reliable user access

6.2. : Password Security:

- Stores passwords in encrypted (hashed) format
- Prevents direct access to user passwords
- Ensures secure storage in the database
- Protects user data from potential breaches
- Follows best practices for data protection

6.3. Access Control:

- Allows only authorized users to access specific features
- Restricts sensitive operations to admin users
- Implements role-based permissions
- Prevents unauthorized modifications to system data
- Ensures controlled and secure system usage

7. User Interface Module:

This module focuses on providing an attractive, user-friendly, and responsive interface to enhance user experience. It ensures smooth interaction between the user and the system.

- Responsive and adaptive design
- Easy navigation across pages
- Clean and structured layout
- Interactive and user-friendly components

7.1. Responsive Design:

The system adapts to different devices

- Adapts to different screen sizes and devices
- Provides mobile-friendly interface
- Supports tablets and desktops efficiently
- Ensures consistent user experience across platforms
- Improves accessibility and usability

7.2. Navigation:

- Provides menu-based navigation system
- Enables easy browsing of products
- Offers quick access to important pages
- Enhances user interaction and convenience
- Improves overall usability of the system

7.3. Forms:

- Includes login and registration forms
- Allows users to input required data easily
- Ensures proper validation of user inputs
- Provides feedback for incorrect or missing data
- Enhances secure and smooth user interaction

8.Checkout/Order Module:

This module handles the final stage of the purchasing process. It ensures that selected products are confirmed, processed, and stored as orders in the system.

- Manages order placement and confirmation
- Provides order summary before final submission
- Ensures accurate and secure transaction handling
- Supports future integration of online payment systems

8.1. Checkout

- Allows users to review selected products before placing an order
- Displays product details such as name, quantity, and total price
- Ensures user confirmation before final submission
- Prevents accidental or incorrect orders
- Improves reliability in the purchasing process

8.2. Order Confirmation:

- Displays complete order summary after placement
- Shows success message to confirm order completion
- Stores order details securely in the database
- Generates confirmation for user reference
- Enhances user trust and transparency

8.3 Future Scope:

- Integration of online payment methods such as UPI and debit/credit cards
- Implementation of secure payment gateways (Razorpay, Stripe, etc.)
- Support for digital wallets and net banking
- Real-time transaction processing and verification
- Expansion to advanced billing and invoice generation

9. Notification Module:

This module provides real-time updates and alerts to users regarding system activities. It improves communication and keeps users informed about important events.

- Sends alerts and notifications to users
- Provides status updates for user actions
- Enhances user engagement and interaction
- Improves transparency in system operations

9.1. User Notification

- Displays login success or failure messages
- Confirms successful user registration
- Provides updates on order status
- Notifies users about important actions and changes
- Ensures better communication between system and user

10.Performance module:

This module focuses on improving system efficiency and ensuring smooth operation under different conditions. It enhances speed, reliability, and resource utilization.

- Ensures fast loading of web pages
- Optimizes database queries for better performance
- Manages system resources efficiently
- Provides stable and responsive user experience

11.Platform-Specific Entry Points:

The system operates across two main platforms: backend and frontend. Each platform has its own entry points to handle user interaction and data processing.

11.1.Backend Entry Point(Django):

The backend serves as the core processingunit of the system.

- Handled business logic

- Processes user requests
- Interacts with database
- Manages authentication

It ensures smooth functioning of all backend operations.

11.2.Frontend Entry Point(Web Interface):

The frontend provides the user interface of the system.

- Built using HTML,CSS,JavaScript
- Handles user input
- Displays outputs
- Sends requests to backend

It acts as the interaction layer between user and system.

11.3.Request-Response Flow:

This defines communication between frontend and backend,

1. User performs action(login,search,add to cart)
2. Request sent to backend
3. Backend process data
4. Response returned to frontend
5. Result displayed to user

11.4.Integration Between Platforms:

The frontend and backend work together to ensure seamless operation.

- Data exchange using HTTP requests.
- Real-time updates
- Smooth user experienc

Data Management and processing:

Data management and processing are essential components of the FarmHub system. This section focuses on how data is collected, validated, stored, processed, and retrieved efficiently. Proper data handling ensures system reliability, security, and smooth user experience.

1. Dataset Collection:

The system collects data from multiple sources such as user inputs and admin entries. This data is dynamically updated and stored in the database.

- User registration data (username, password)
- Product details (name, price, description, images)
- Cart information (selected items, quantity)
- Order details (user orders, purchase history)

The collected data is structured and organized to support efficient processing and retrieval.

2. Data Validation :

Before storing data in the database, it is validated to ensure accuracy and consistency.

- Required field validation
- Duplicate username checking
- Proper data formats
- Removal of invalid inputs

This step helps prevent errors and ensures reliable system performance.

3. Data Storage:

Data is processed and stored in a structured format suitable for efficient retrieval.

- Organized database tables
- Use of Django models
- Proper field types (CharField, IntegerField, etc.)
- Relationship handling (User–Order, Product–Cart)

This ensures smooth interaction between different modules of the system.

4. Data Processing:

Data is processed based on user actions and system requirements.

- Processing login credentials
- Filtering products based on search
- Calculating total price in cart
- Managing order placement

This ensures that the system responds correctly to user requests.

5.Data Retrievel

Efficient retrieval mechanisms are used to fetch data from the database.

- Fetching product lists
- Displaying user-specific data
- Retrieving order history
- Search-based filtering

Optimized queries ensure faster response time and better performance.

6.Image Handling

The system manages product images effectively.

- Image upload functionality
- Storage in media directory
- Supported formats (JPG, PNG,Links)
- Display optimization

This improves visual representation and user experience.

7.Data Flow Management:

The system ensures smooth data flow between frontend and backend.

- User enters data
- Data is validated
- Sent to backend
- Stored in database
- Retrieved and displayed

This structured flow ensures consistency and reliability.

Database Design:

The database design of the FarmHub system is structured to efficiently store and manage data related to users, products, carts, and orders. A relational database model is used to maintain proper relationships between different entities and ensure data consistency.

1. **User Table:**

This table stores information about registered users.

- Username
- Password (hashed)
- User

Each user is uniquely identified and authenticated using this data.

2. Product Table:

This table contains details of all farm products available on the platform.

- Product ID
- Product Name
- Price
- Description
- Image

It helps in displaying and managing products effectively.

3. Cart Table:

It helps in displaying and managing products effectively.

- Cart ID
- User (Foreign Key)
- Product (Foreign Key)
- Quantity

It allows users to manage items before placing an order.

4. Order Table:

This table stores completed order details.

- Order ID
- User (Foreign Key)
- Products
- Total Amount
- Order Date

It maintains a record of all user purchases.

5. **Relationships Between Tables:**

The tables are connected using relationships to ensure proper data flow.

- One user → Multiple orders
- One user → Multiple cart items
- One product → Multiple cart entries

These relationships help maintain

6. **Database Operations:**

The system uses Django ORM to perform database operations.

- Create (Insert data)
- Read (Fetch data)
- Update (Modify data)
- Delete (Remove data)

This simplifies interaction with the database without writing complex SQL queries

Proposed System Workflow:

The proposed system workflow of FarmHub – An Online Farm Website illustrates the complete operational flow of the application, describing how users interact with the system and how data is processed between the frontend and backend. The workflow ensures smooth navigation, efficient processing, secure transactions, and an enhanced user experience.

1) User Registration & Login:

The workflow begins with user authentication, which ensures that only authorized users can access system features.

i. User visits the website through a browser
ii. New users register by providing username, email, and password
iii. Existing users log in using valid credentials
iv. The system verifies user details using authentication mechanisms
v. Passwords are securely stored using encryption techniques

vi. Upon successful authentication, access is granted to the system

This step ensures security, user validation, and controlled access

2) Product Browsing:

After successful login, users can explore available farm products.

i. Displays a list of products stored in the database

ii. Users can search products by name or category

iii. Products are organized into categories (fruits, vegetables, grains, etc.)

iv. Pagination is used for handling large product lists

v. Users can view detailed product information including price, description, and image

This improves user experience and easy product discovery

3) Cart Operations:

The cart module allows users to manage selected products before purchasing.

i. Users can add products to the cart

ii. Modify quantity of selected items

iii. Remove unwanted products from cart

iv. System calculates total price dynamically

v. Cart acts as temporary storage before order confirmation

Ensures flexibility and convenience in shopping

4) Order Placement:

Users review selected items in cart

- Confirm order details
- Order is placed successfully
- Order data is stored in the database
- Unique order ID is generated for tracking

This completes the transaction process

5) Backend Processing :

The backend plays a crucial role in handling all system operations and business logic. It ensures that user requests are processed accurately and efficiently.

i. Receives user requests from the frontend through HTTP methods

ii. Validates input data to ensure correctness and prevent errors

iii. Implements business logic for operations such as login, product retrieval, cart updates, and order placement

iv. Interacts with the database using Django ORM for data storage and retrieval

v. Processes orders and updates system records in real-time

vi. Ensures secure handling of sensitive information

This guarantees efficient processing, data consistency, and system reliability

6) Result Display :

Once the backend processes the request, the results are dynamically displayed to the user through the frontend interface.

- Product details are rendered using templates
- Updated cart information is displayed instantly
- Order confirmation messages are shown after successful transactions
- User-friendly messages are provided for actions like login success or errors
- Dynamic content is displayed using Django template engine

This ensures clear communication and better user experience

7)Data Flow Sequence:

The system follows a structured data flow to maintain smooth communication between components:

1. User sends request (login, search, add to cart, etc.)
2. Frontend captures input and forwards request via URL routing
3. Django URL dispatcher maps request to appropriate view
4. Backend (views) processes logic

5. Models interact with database (store/retrieve/update data)
6. Response is generated and sent back to frontend
7. Output is rendered using HTML templates and displayed to user

This structured flow ensures efficient frontend-backend integration

8)Workflow Advantages:

The proposed workflow offers several advantages:

- Smooth and intuitive user interaction
- Fast and efficient data processing using Django framework
- Secure authentication and data handling
- Scalable architecture for future enhancements
- Modular design for easy maintenance and updates
- Reliable database management using ORM

These advantages contribute to high performance and system stability

Implementation Details:

The implementation of the FarmHub – An Online Farm Website involves the integration of frontend design, backend logic, and database management. The system is developed using Django as the backend framework and HTML, CSS, and JavaScript for the frontend interface.

1. Development Environment:

The project is developed using the following tools and environment:

- ✓ Programming Language: Python
- ✓ Framework: Django
- ✓ Frontend: HTML, CSS, JavaScript
- ✓ Database: SQLite (default Django database)
- ✓ IDE: Visual Studio Code
- ✓ Platform: Web Browser

This environment provides a flexible and efficient platform for web application development.

2. Frontend Implementation :

The frontend is responsible for user interaction and interface design.

Features:

- ✓ Login and Registration pages
- ✓ Product listing page
- ✓ Cart interface
- ✓ Order confirmation page

Technologies:

- ✓ HTML for structure
- ✓ CSS for styling
- ✓ JavaScript for interactivity

The frontend ensures a clean, responsive, and user-friendly experience

3. Backend Implementation(Django):

The backend handles application logic and data processing.

Key Functionalities:

- ◆ User authentication (login, logout, register)
- ◆ Product management
- ◆ Cart operations
- ◆ Order processing.

Implementation:

- ➢ Views (views.py) handle user requests
- ➢ URLs (urls.py) manage routing
- ➢ Models (models.py) define database structure

Django ensures secure and scalable backend operations.

4. Database Implementation:

The database stores all application data.

Tables Used:

- User

- Product
- Cart
- Order

Features:

- ✓ Data stored using Django ORM
- ✓ Relationships managed using Foreign Keys
- ✓ Automatic database handling using migrations

5. Authentication System:

The system uses Django's built-in authentication mechanism.

- User registration using create_user()
- Login using authenticate()
- Session handling using login()
- Logout functionality

Passwords are stored securely using hashing.

6. Cart and Order Logic:

The cart and order system is implemented to manage purchases.

Cart:

- Add products
- Update quantity
- Remove items

Order:

- Convert cart to order
- Store order details
- Display order history

7.Search and Pagination:

These features improve usability and performance.

1. Search using query filters
2. Pagination using Django Paginator

3. Efficient data loading

8.Error Handling and Validation:

The system handles errors and invalid inputs effectively.

1. Invalid login message
2. Duplicate username prevention
3. Required field validation
4. Exception handling

9.Integration of Components:

All components are integrated to ensure smooth workflow.

1. Frontend sends request
2. Backend processes logic
3. Database stores/retrieves data
4. Response displayed to user.

Challenges Faced & Solutions :

During the development of the FarmHub – An Online Farm Website, several challenges were encountered in both frontend and backend implementation. These challenges helped in gaining practical experience and improving problem-solving skills. The following are the major challenges faced and the solutions implemented.

1. User Authentication Issues:

Challenge:

Users were unable to log in even after registration due to incorrect handling of user creation and authentication.

Solution:

Used Django's create_user() method to properly store hashed passwords.

Implemented authenticate() and login() functions correctly.

Fixed incorrect imports (User model).

2. Database Migration Errors:

Challenge:

Errors occurred while modifying models, especially when adding new fields like description or image.

Solution:

Used makemigrations and migrate properly.

Provided default values when required.

Updated models carefully.

3. Image Display Issues:

Challenge:

Product images were not displaying correctly on the website.

Solution:

Configured media settings (MEDIA_URL, MEDIA_ROOT).

Used correct image URLs.

Ensured proper file paths.

4. Cart Functionality Problems:

Challenge:

Cart was not updating correctly when adding or removing items.

Solution:

Fixed cart logic in backend.

Used proper relationships between user and product.

Implemented quantity update correctly.

5.Page Rounting Errors:

Challenge:

Errors like “Page Not Found (404)” occurred during navigation.

Solution:

Corrected URL patterns in urls.py.

Ensured proper linking between pages.

Verified route names.

6. UI Design Issues:

Challenge: The interface initially looked unprofessional and inconsistent.

Solution:

Improved CSS styling.

Used proper layout design.

Adjusted spacing, alignment, and colors.

7.Server and Execution Issues:

Challenge:

Problems occurred while running the server and activating environments.

Solution:

Fixed virtual environment setup.

Restarted server after changes.

Verified Python and Django installation.

FUTURE SCOPE:

The FarmHub – An Online Farm Website can be further enhanced with advanced features and technologies to improve functionality, scalability, performance, and user experience. The following enhancements can be implemented in future versions of the system:

1. Online Payment Integration:

Currently, the system supports order placement without integrated payment functionality.

- Integration with secure payment gateways such as UPI, Debit/Credit Cards, and Net Banking
- Real-time payment processing and confirmation
- Secure transaction handling using encryption protocols
- Automatic invoice generation after successful payment
- Refund and cancellation management

This will transform the system into a complete e-commerce platform

2. Delivery Tracking System:

A delivery tracking module can improve transparency and user satisfaction.

- Estimated delivery time calculation
- Integration with GPS-based tracking systems
- Notifications for each stage of delivery
- Delivery partner management system

Ensures better customer experience and trust

3. Mobile Application Development:

To increase accessibility, the system can be extended to mobile platforms.

- Development of Android and iOS applications
- Cross-platform support using frameworks like Flutter/React Native
- Push notifications for order updates and offers
- Offline browsing capabilities
- Improved user engagement through mobile-friendly UI

Provides anytime, anywhere access

4. Advanced Search and Filters:

Enhancing search functionality will improve usability and efficiency.

- Filter products based on price, category, ratings, and availability
- Intelligent search suggestions using auto-complete
- Sorting options (price low to high, popularity, newest)
- Voice-based search integration
- AI-based product recommendation during search

Enables quick and accurate product discovery

5.User Profile Enhancement:

User accounts can be improved for personalization and convenience.

- Maintain order history and transaction records

- Save multiple delivery addresses
- Profile customization options
- Wishlist feature for saving favorite products
- Personalized dashboard for users

Improves user engagement and personalization

6.Recommendation System:

A smart recommendation system can enhance user experience.

- Suggest products based on browsing and purchase history
- Use machine learning algorithms for personalized recommendations
- Display trending and popular products
- Offer discounts based on user preferences

Provides a personalized shopping experience

7.Multi-Vector Support:

The platform can be expanded to support multiple sellers (farmers/vendors).

- Farmers can register and upload their products
- Separate seller dashboards for product management
- Vendor-wise product categorization
- Commission-based earning model
- Performance analytics for vendors

Converts system into a marketplace platform

8.Security Enhancements:

Advanced security mechanisms can be implemented to protect user data.

- Two-factor authentication (2FA) for login
- Advanced password encryption techniques
- Secure payment verification systems

- Protection against SQL injection and cyber attacks
- Regular data backups and recovery mechanisms

Ensures data privacy and system security

9.Notification System:

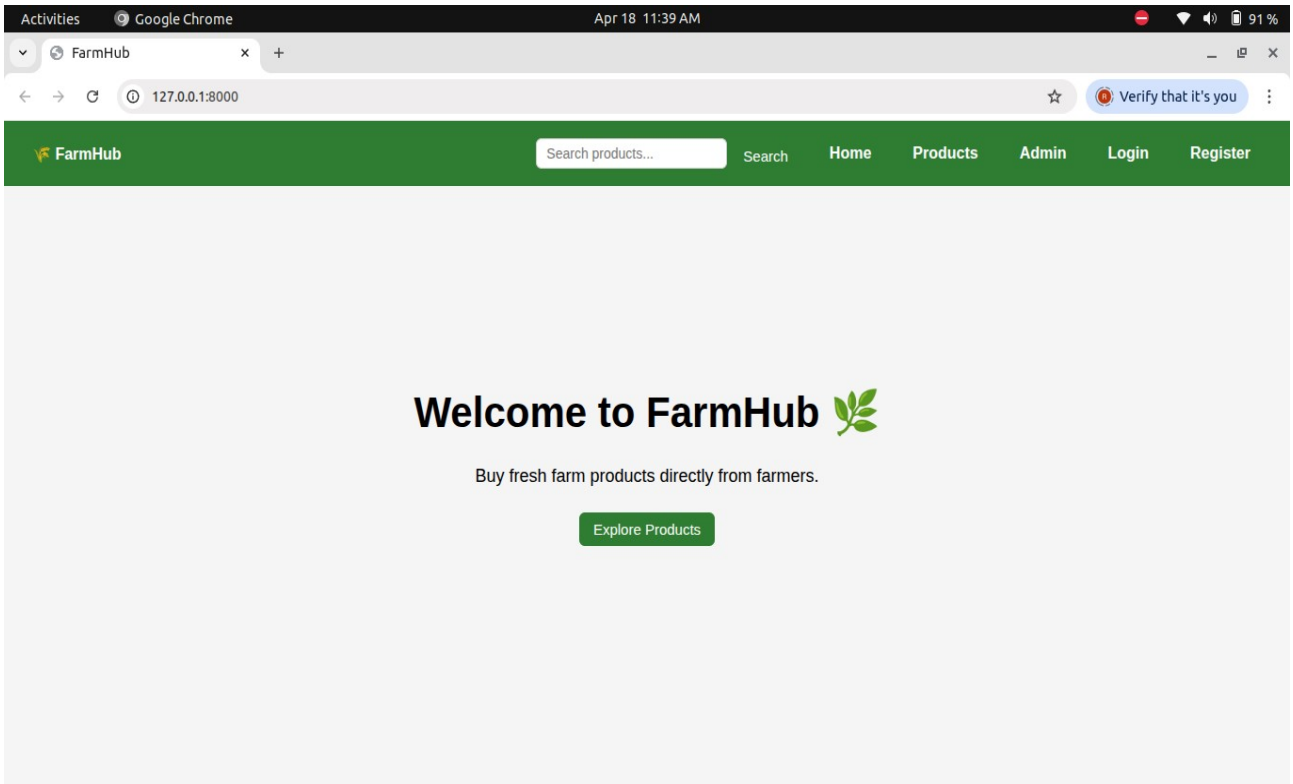
A real-time notification system can be added.

- Email and SMS notifications for order updates
- Push notifications for offers and discounts
- Alerts for delivery status changes
- Reminder notifications for pending actions

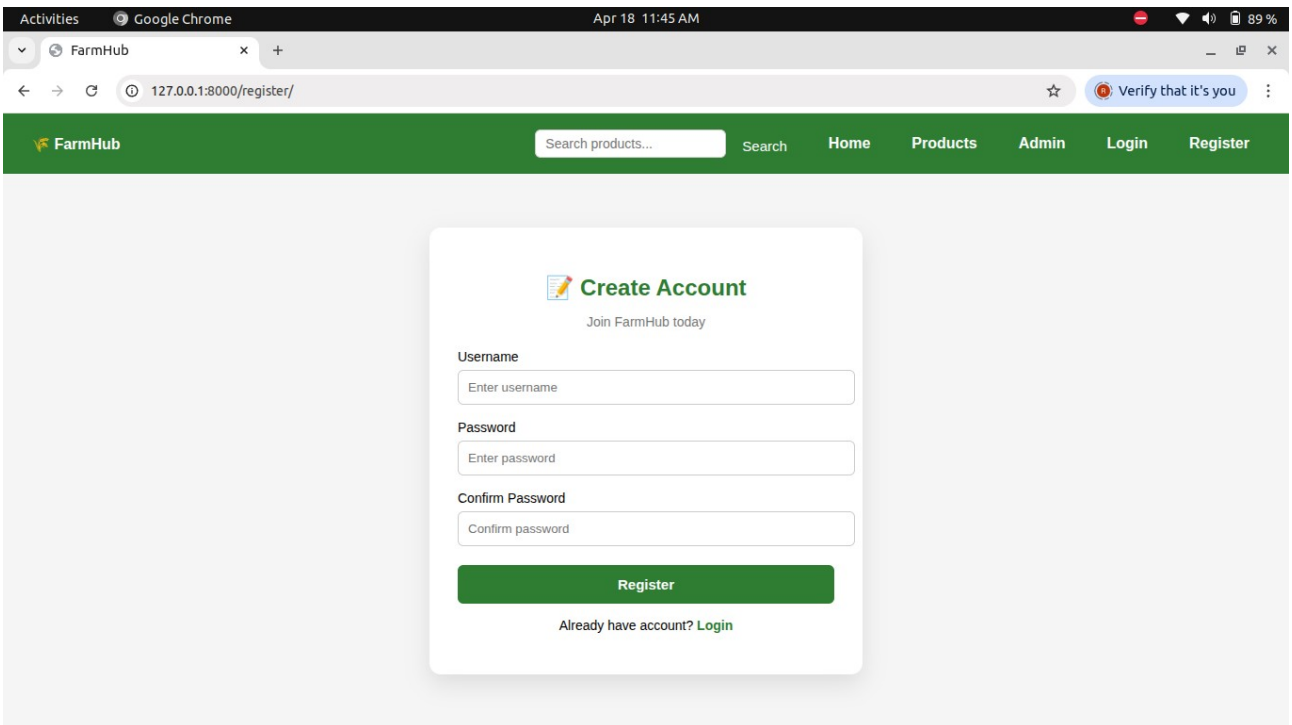
Improves communication and user engagement

Farmhub Website

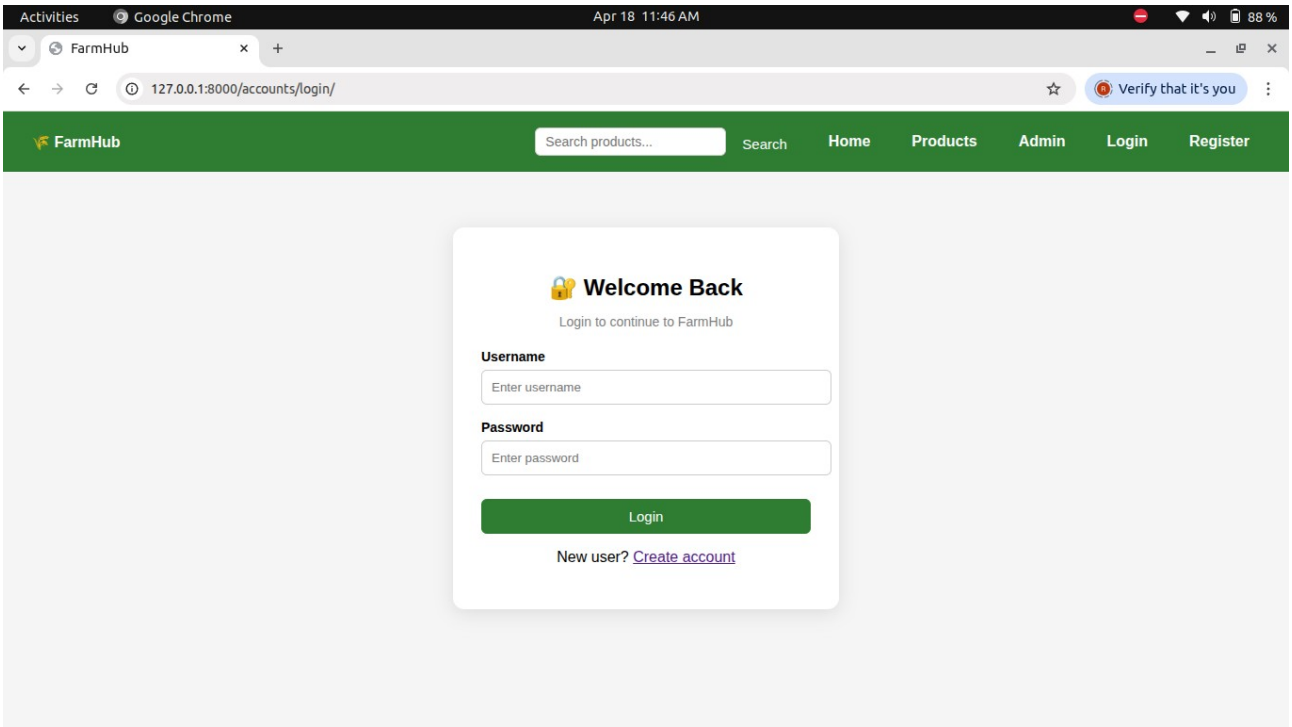
Home Page:



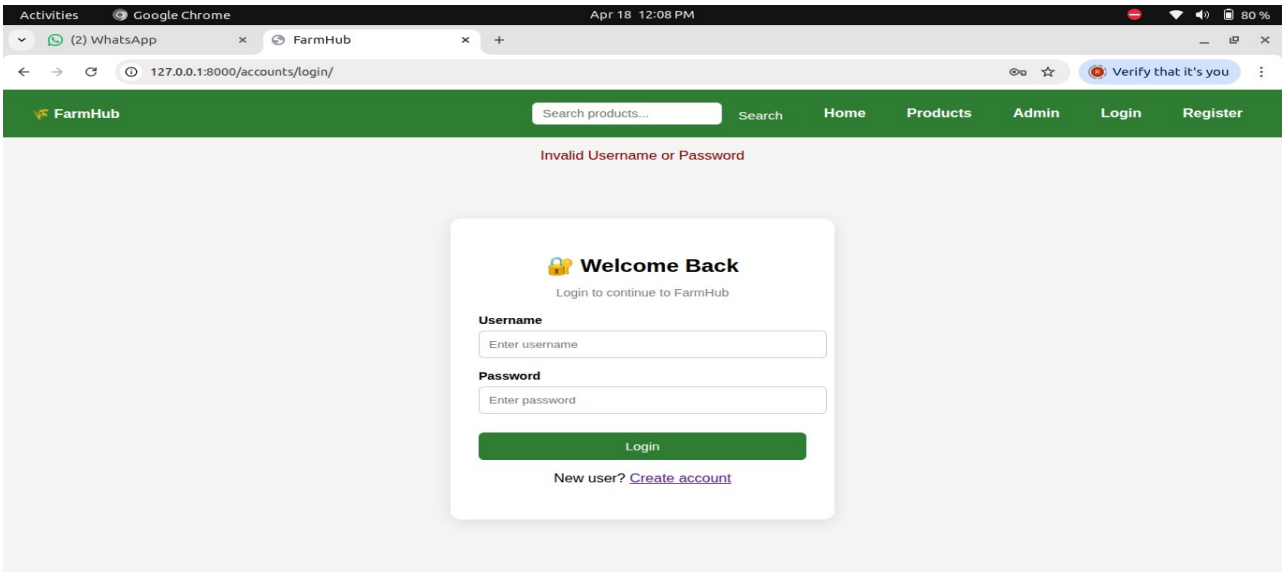
Registration Page:



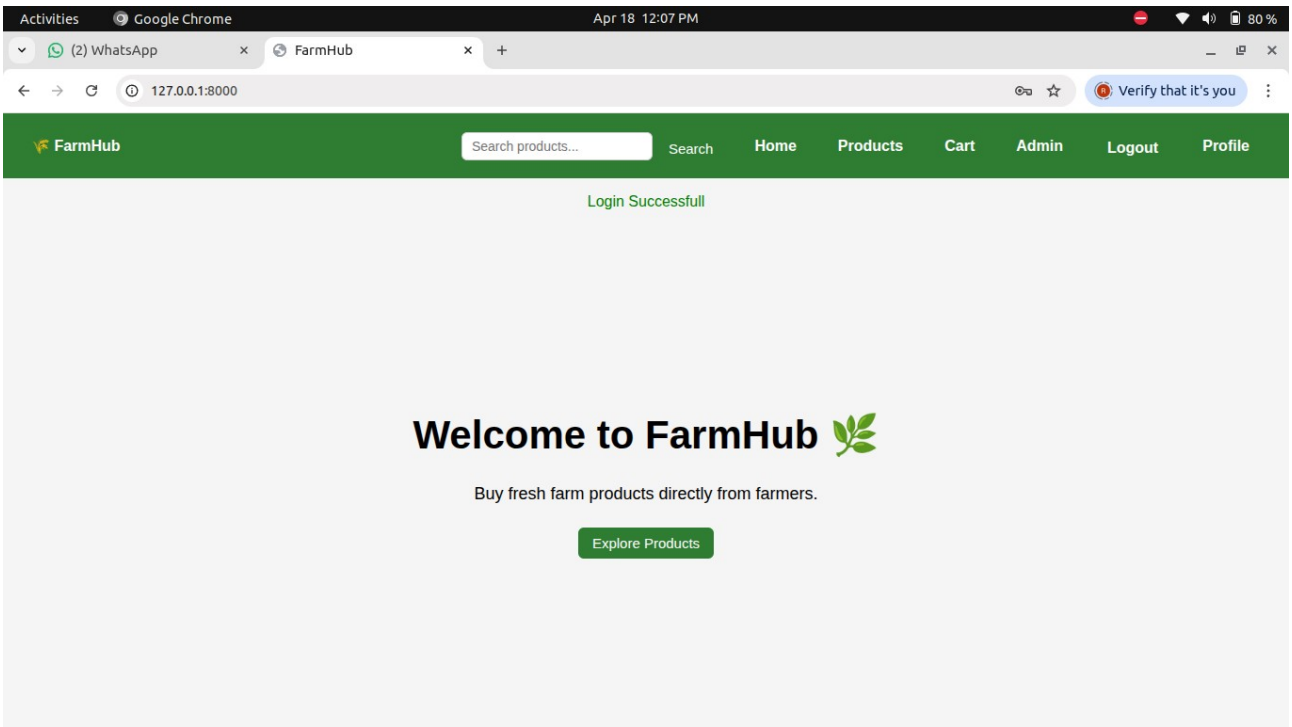
Login Page:



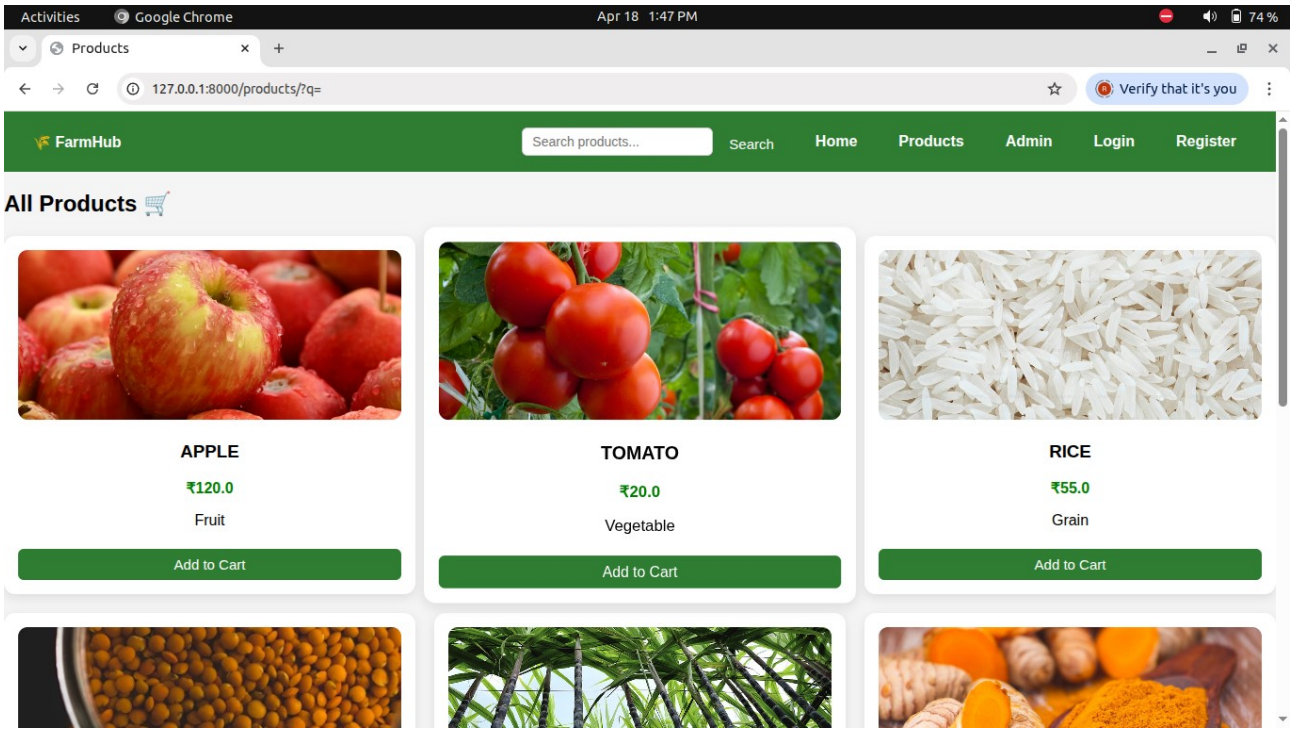
Login Failure:

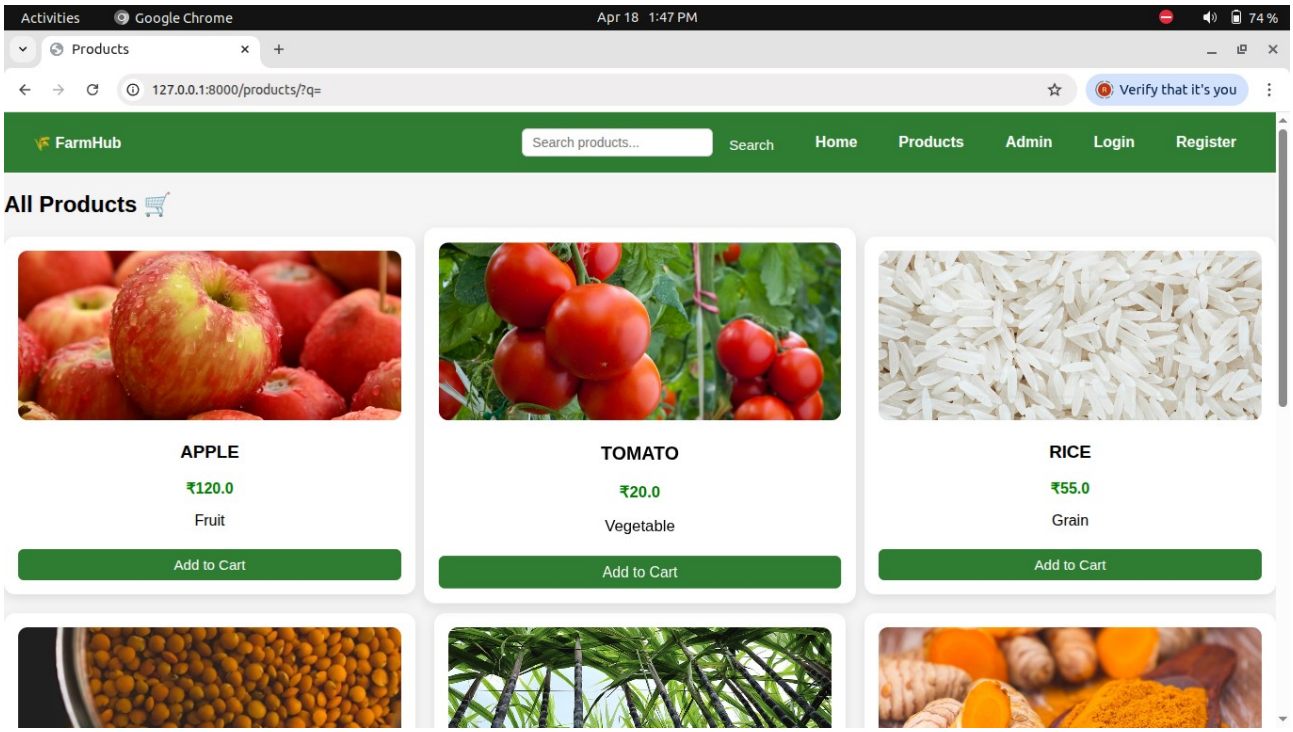


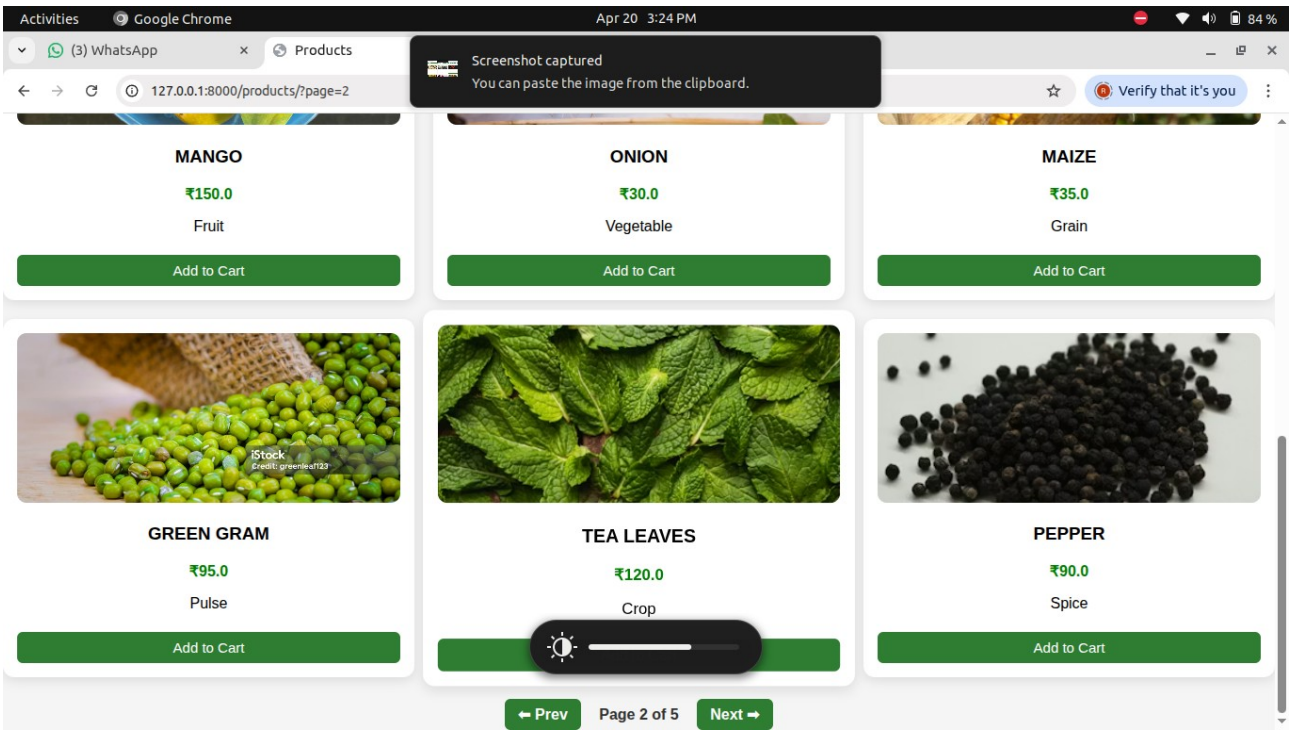
Login Success:

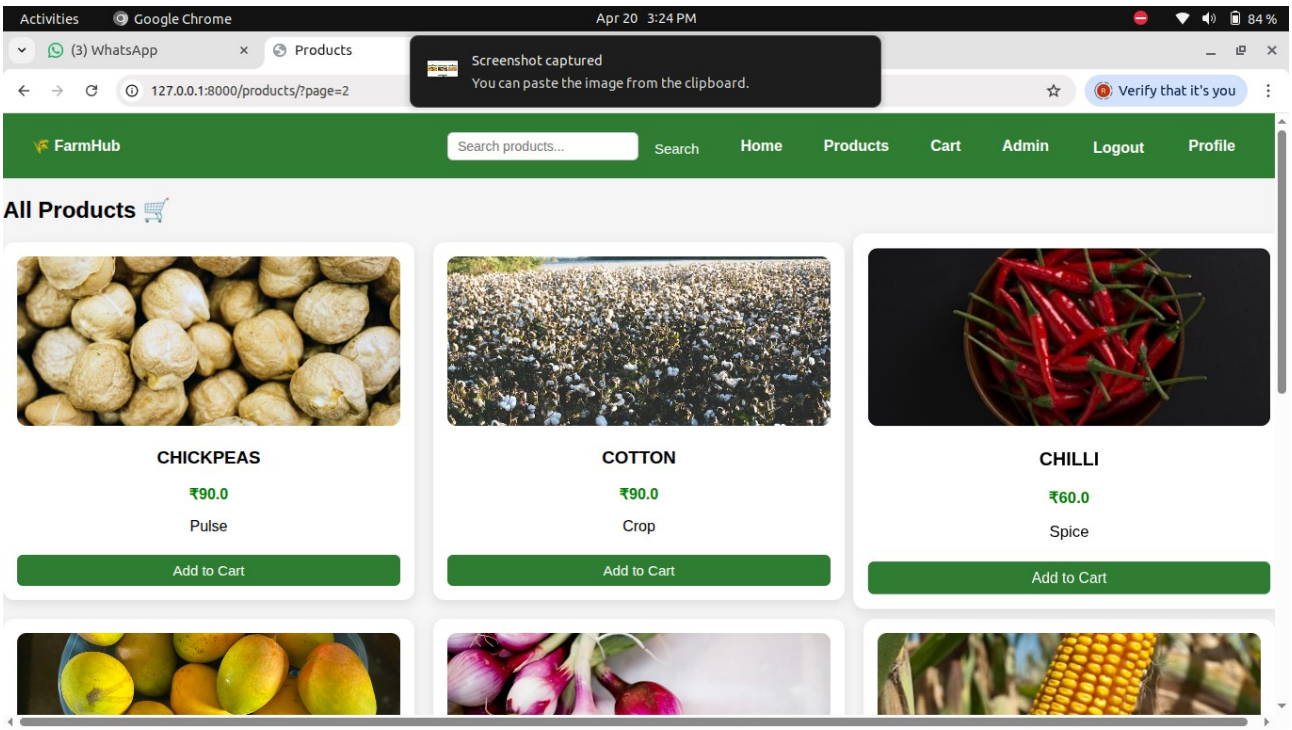


Product Page:




Activities
Google Chrome
Apr 18 1:47 PM
74 %
Products
127.0.0.1:8000/products/?q=
Verify that it's you
FarmHub
Search products...
Search
Home
Products
Admin
Login
Register
All Products
APPLE
₹120.0
Fruit
Add to Cart
TOMATO
₹20.0
Vegetable
Add to Cart
RICE
₹55.0
Grain
Add to Cart


Activities
Google Chrome
Apr 20 3:24 PM
84 %
(3) WhatsApp
Products
127.0.0.1:8000/products/?page=2
Screenshot captured
You can paste the image from the clipboard.
Verify that it's you
MANGO
₹150.0
Fruit
Add to Cart
ONION
₹30.0
Vegetable
Add to Cart
MAIZE
₹35.0
Grain
Add to Cart
GREEN GRAM
₹95.0
Pulse
Add to Cart
TEA LEAVES
₹120.0
Crop
PEPPER
₹90.0
Spice
Add to Cart
Prev
Page 2 of 5
Next


Activities
Google Chrome
Apr 20 3:24 PM
84 %
(3) WhatsApp
Products
Screenshot captured
You can paste the image from the clipboard.
127.0.0.1:8000/products/?page=2
Verify that it's you
FarmHub
Search products...
Search
Home
Products
Cart
Admin
Logout
Profile
All Products
CHICKPEAS
₹90.0
Pulse
Add to Cart
COTTON
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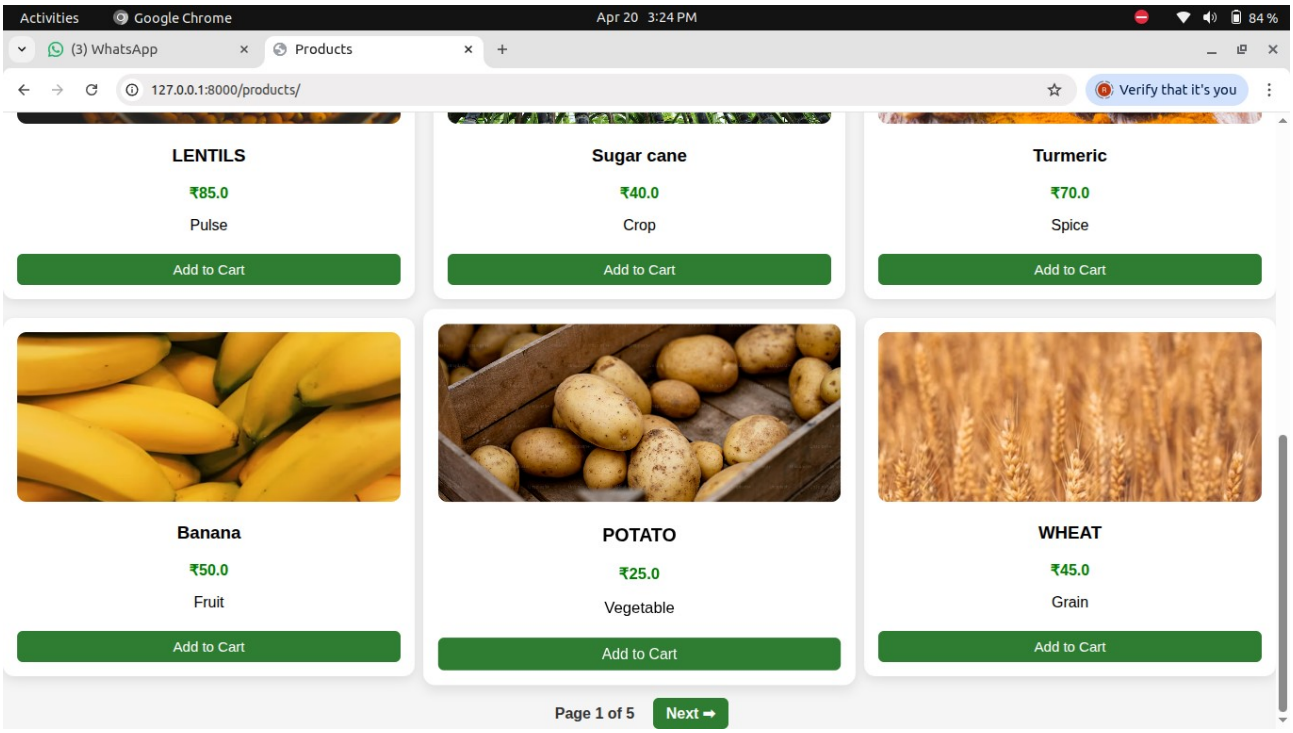
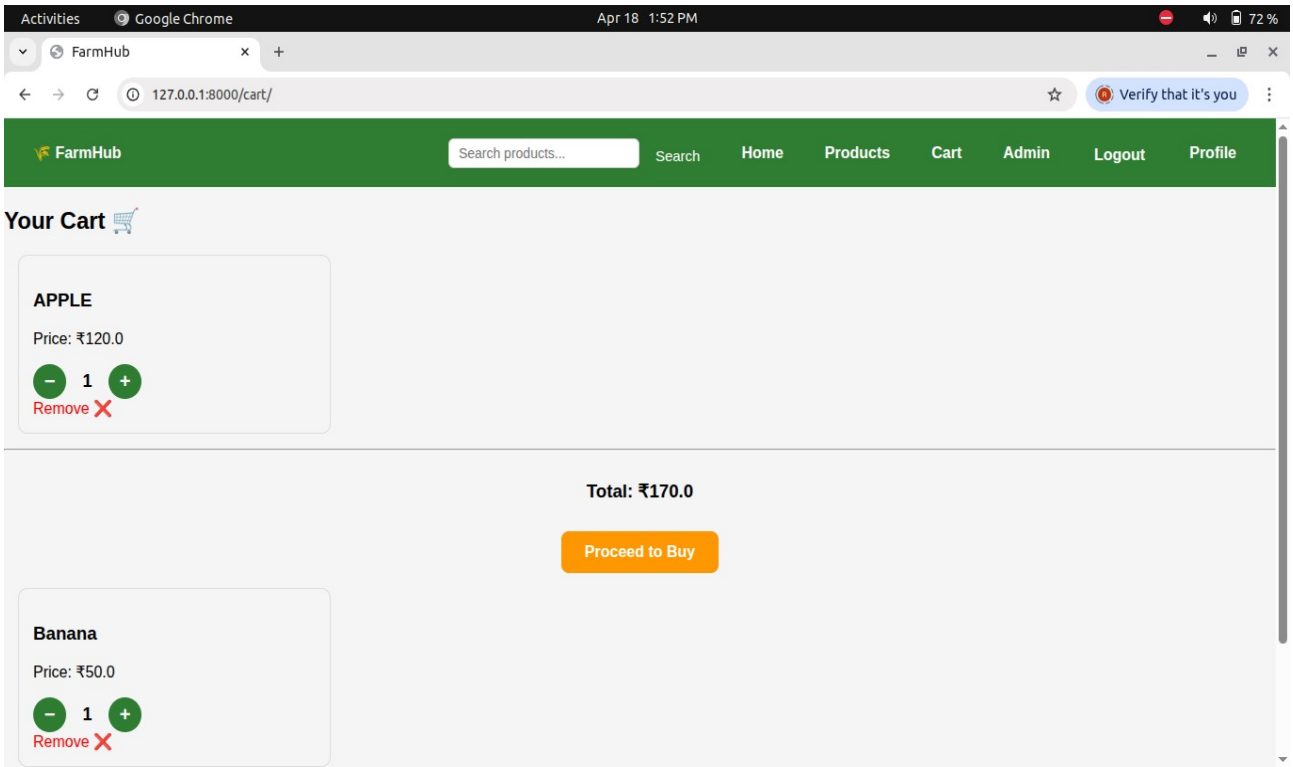
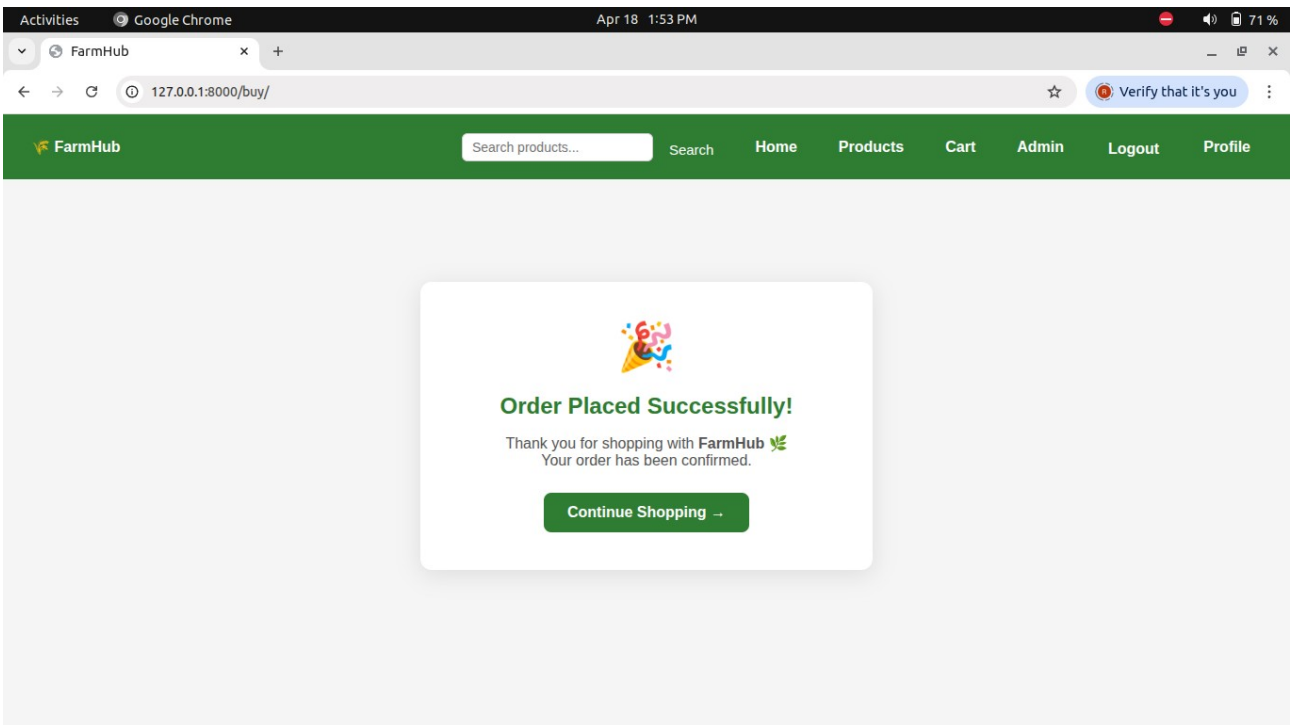
Crop
Add to Cart
CHILLI
₹60.0
Spice
Add to Cart


Activities
Google Chrome
Apr 20 3:24 PM
84 %
(3) WhatsApp
Products
127.0.0.1:8000/products/
Verify that it's you
LENTILS
₹85.0
Pulse
Add to Cart
Sugar cane
₹40.0
Crop
Add to Cart
Turmeric
₹70.0
Spice
Add to Cart
Banana
₹50.0
Fruit
Add to Cart
POTATO
₹25.0
Vegetable
Add to Cart
WHEAT
₹45.0
Grain
Add to Cart
Page 1 of 5
Next ➡

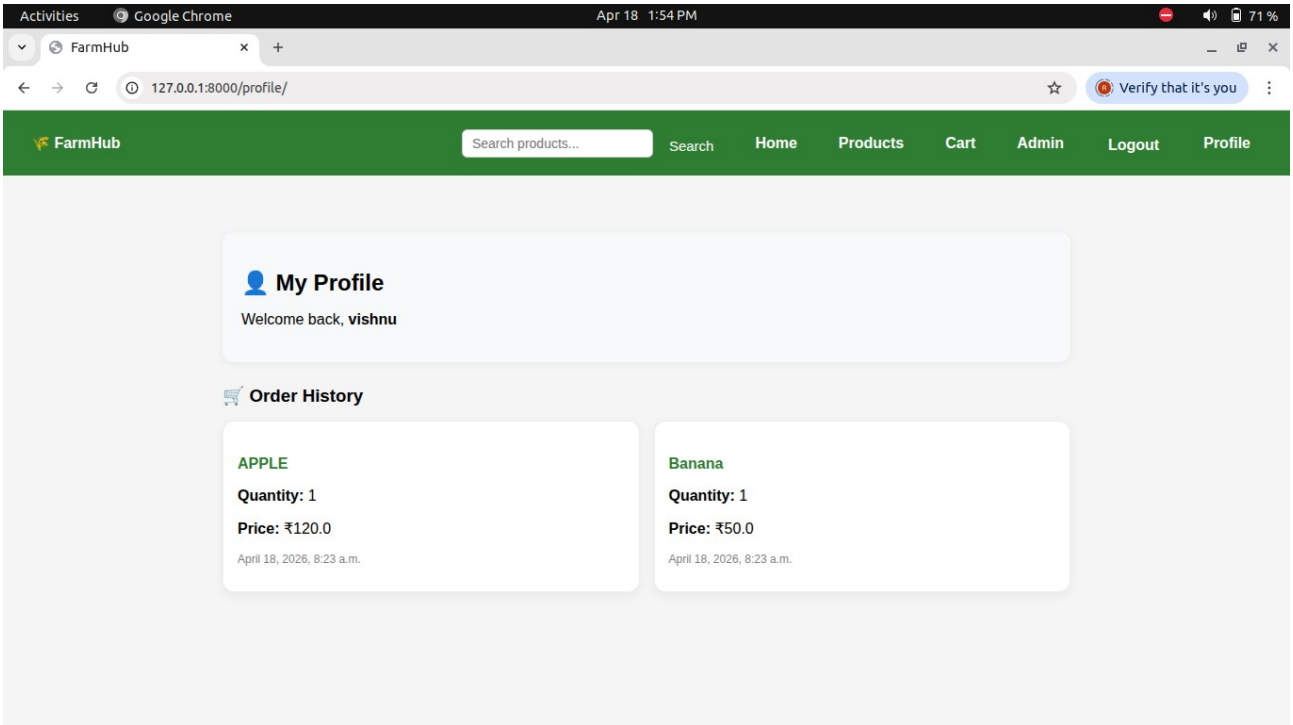
Cart Page:



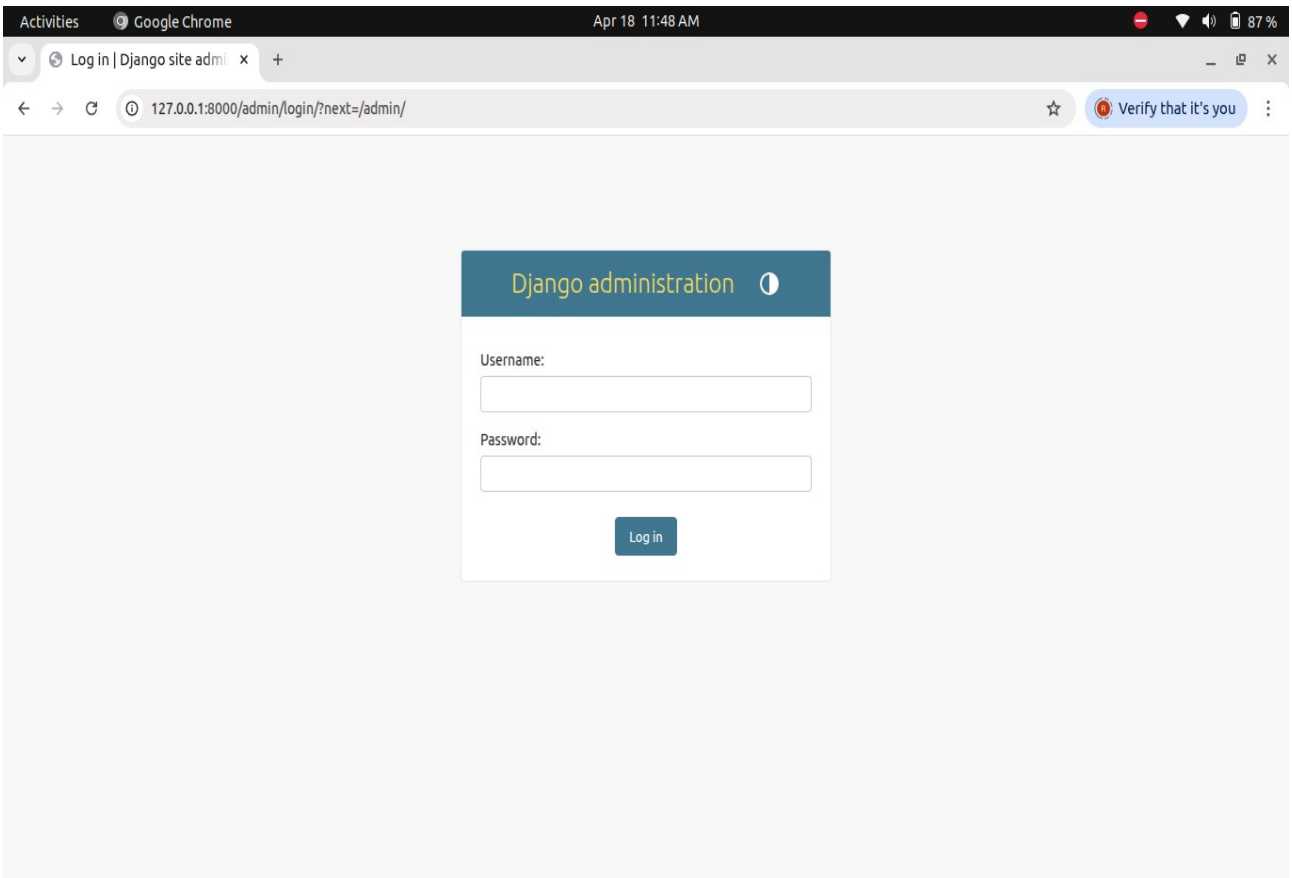
Order Success:



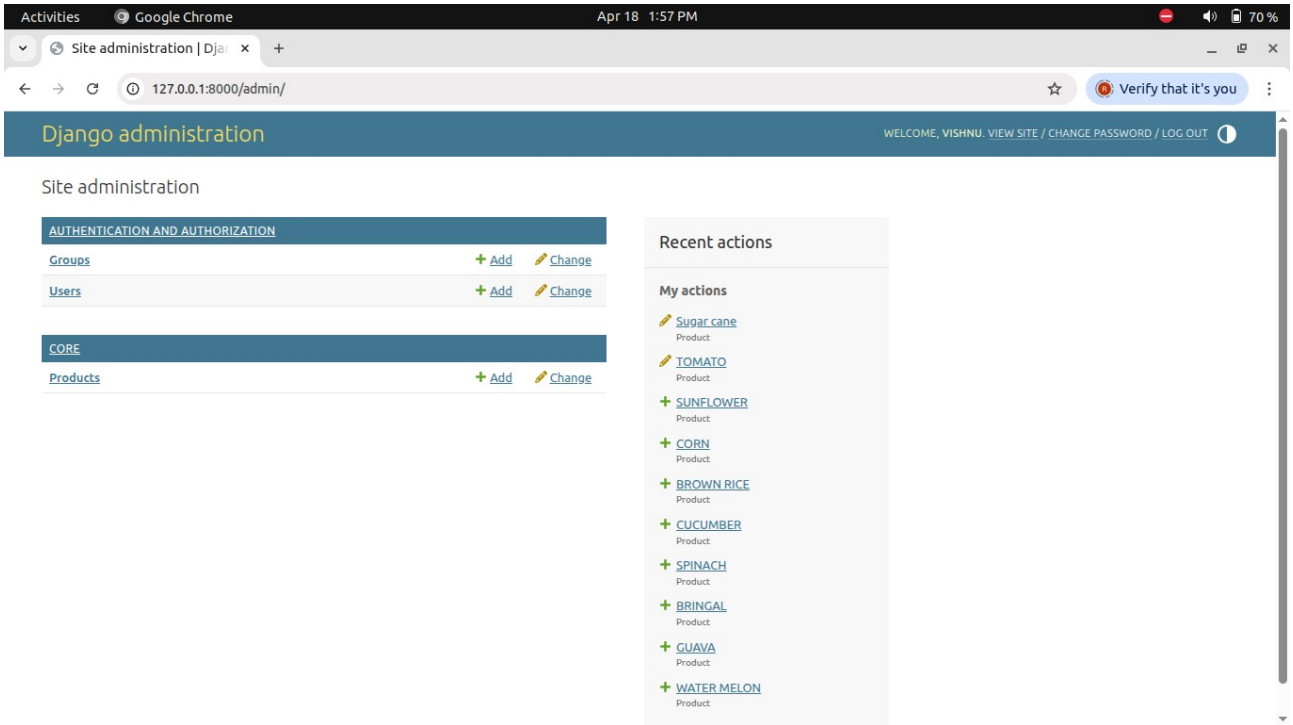
Profile:



Django Admin Login Interface:



Django Admin Dashboard:



Implementation of Product model:

```
class Product(models.Model):

    CATEGORY_CHOICES = [
        ('Fruit', 'Fruit'),
        ('Vegetable', 'Vegetable'),
        ('Grain', 'Grain'),
        ('Pulse', 'Pulse'),
        ('Crop', 'Crop'),
        ('Spice', 'Spice'),
    ]

    name = models.CharField(max_length=100)
    price = models.FloatField()
    category = models.CharField(max_length=20, choices=CATEGORY_CHOICES)
    description = models.TextField()
    image = models.URLField(max_length=500, null=True, blank=True)
```

Proect URL configuration:

```
from django.contrib import admin
from django.urls import path,include
from core import views

urlpatterns = [
    path("admin/", admin.site.urls),
    path("",include("core.urls")),
    path("accounts/",include("django.contrib.auth.urls")),
    path("buy/",views.buy_view),
]
```

URL Configuration for Core Application:

```
from django.urls import path
from . import views


urlpatterns = [
    path('', views.home, name='home'),
    path('products/', views.products, name='products'),
    path('register/',views.register,name='register'),
    path('accounts/login/', views.user_login, name='login'),
    path('add-to-cart/<int:product_id>/', views.add_to_cart, name='add_to_cart'),
    path('cart/',views.cart_view,name='cart'),
    path('increase/<int:item_id>/', views.increase_quantity),
    path('decrease/<int:item_id>/', views.decrease_quantity),
    path('remove/<int:item_id>/', views.remove_from_cart),
    path('profile/', views.profile, name='profile'),
]
```

Views(Businness Logic Implementation):

```
from django.shortcuts import render, redirect
from .models import Product

def home(request):
    products = Product.objects.all()
    return render(request, 'home.html', {'products': products})


def buy_view(request):
    return render(request, 'buy.html')
```

Frontend Templates(User Interface):

```html
<!DOCTYPE html>
<html>
<head>
    <title>FarmHub</title>
</head>
<body>
{% extends 'base.html' %}

{% block content %}

<div class="hero">

    <h1>Welcome to FarmHub 🌿</h1>
    <p>Buy fresh farm products directly from farmers.</p>

    <a href="/products/">
        <button>Explore Products</button>
    </a>

</div>
{% if user.is_authenticated and user.is_superuser %}


{% endif %}
{% endblock %}

</body>
</html>
```

Purchase Interface:

```html
{% extends 'base.html' %}

{% block content %}

<div class="success-container">
    <div class="success-card">
        <div class="success-icon">🎉</div>

        <h2>Order Placed Successfully!</h2>
        <p class="success-text">
            Thank you for shopping with <b>FarmHub 🌿</b><br>
            Your order has been confirmed.
        </p>

        <a href="/products/" class="continue-btn">
            Continue Shopping →
        </a>
    </div>
</div>

{% endblock %}
```

Admin Configuration:

```python
from django.contrib import admin
from .models import Product

admin.site.register(Product)
```

Conclusion:

The project “FarmHub – An Online Farm Website” has been successfully designed and developed as a full-stack web application that provides an efficient platform for managing and accessing farm products online. The system enables users to register, log in, browse available products, and interact with the application in a simple and user-friendly manner.

The implementation of this project using modern web technologies such as Django, HTML, CSS, and JavaScript has helped in creating a structured and scalable system. The backend, powered by Django, efficiently handles user authentication, data processing, and database interactions, while the frontend ensures a smooth and interactive user experience.

One of the key achievements of this project is the integration of essential modules such as user authentication, product management, and administrative control through the Django Admin Panel. The admin interface provides a powerful and secure way to manage products and users, making the system easy to maintain and extend.

The project also demonstrates the importance of proper system design, including modular architecture, clean user interface design, and efficient data handling. Error handling mechanisms, such as login validation, improve system reliability and user experience

Overall, the FarmHub system fulfills its objective of creating a basic online platform for farm-related product management. It provides a strong foundation for future enhancements such as online payment integration, real-time order tracking, recommendation systems, and mobile application support.

In conclusion, this project not only meets the functional requirements but also provides valuable practical experience in web development, database management, and system design. It serves as a stepping stone toward building more advanced and real-world applications in the field of web technologies.